Testing pytearcat

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Abstract

This document records the result of testing pytearcat[1] against [2]

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1 Test methodology

Take a couple of examples from [2], copy metric, compute Christoffel symbols and Einstein tensor using , and compare with [2]. If a component is zero in both [2] and [1], omit it from report.

2 Schwartzchild Solution

[2][Lecture IV]

2.1 Metric

$$ds^2 = e^{\nu(t,r)} \cdot dt^2 - e^{\lambda(t,r)} \cdot dr^2 - r^2 \cdot d\theta^2 - r^2 \sin^2(\theta) \cdot d\phi^2$$

2.2 Christoffel Symbols

$$\Gamma^0{}_{0\,0} = \frac{\frac{\partial}{\partial t}\nu(t,r)}{2} \qquad \text{Pass}$$

$$\Gamma^0{}_{0\,1} = \frac{\frac{\partial}{\partial r}\nu(t,r)}{2}$$

$$\Gamma^0{}_{1\,0} = \frac{\frac{\partial}{\partial r}\nu(t,r)}{2} \qquad \text{Pass}$$

$$\Gamma^0{}_{1\,1} = \frac{e^{\lambda(t,r)}e^{-\nu(t,r)}\frac{\partial}{\partial t}\lambda(t,r)}{2} \qquad \text{Pass}$$

$$\Gamma^0{}_{0\,0} = \frac{\frac{\partial}{\partial t}\nu(t,r)}{2} \qquad \text{Pass}$$

$$\Gamma^0{}_{0\,1} = \frac{\frac{\partial}{\partial r}\nu(t,r)}{2}$$

$$\Gamma^0{}_{1\,0} = \frac{\frac{\partial}{\partial r}\nu(t,r)}{2} \qquad \text{Pass}$$

$$\Gamma^0{}_{1\,1} = \frac{e^{\lambda(t,r)}e^{-\nu(t,r)}\frac{\partial}{\partial t}\lambda(t,r)}{2} \qquad \text{Pass}$$

$$\Gamma^1{}_{0\,0} = \frac{e^{-\lambda(t,r)}e^{\nu(t,r)}\frac{\partial}{\partial r}\nu(t,r)}{2} \qquad \text{Pass}$$

$$\Gamma^1{}_{0\,1} = \frac{\frac{\partial}{\partial t}\lambda(t,r)}{2} \qquad \text{Pass}$$

$$\Gamma^1{}_{1\,0} = \frac{\frac{\partial}{\partial t}\lambda(t,r)}{2}$$

$$\Gamma^1{}_{1\,1} = \frac{\frac{\partial}{\partial r}\lambda(t,r)}{2} \qquad \text{Pass}$$

$$\Gamma^1{}_{2\,2} = -\,re^{-\lambda(t,r)} \qquad \text{Pass}$$

$$\Gamma^1{}_{3\,3} = -\,re^{-\lambda(t,r)}\sin^2(\theta) \qquad \text{Pass}$$

$$\Gamma^2{}_{1\,2} = \frac{1}{r} \qquad \text{Pass}$$

$$\Gamma^2{}_{2\,1} = \frac{1}{r}$$

$$\Gamma^2{}_{3\,3} = -\sin(\theta)\cos(\theta) \qquad \text{Pass}$$

$$\Gamma^3{}_{1\,3} = \frac{1}{r} \qquad \text{Pass}$$

$$\Gamma^3{}_{2\,3} = \frac{\cos(\theta)}{\sin(\theta)}$$

$$\Gamma^3{}_{3\,1} = \frac{1}{r}$$

$$\Gamma^3{}_{3\,2} = \frac{\cos(\theta)}{\sin(\theta)} \qquad \text{Pass}$$

2.3 Einstein Tensor

Results from pytearcat.

$$G_{0\,0} = \frac{\left(r\frac{\partial}{\partial r}\lambda(t,r) + e^{\lambda(t,r)} - 1\right)e^{-\lambda(t,r)}e^{\nu(t,r)}}{r^2}$$

$$G_{0\,1} = \frac{\frac{\partial}{\partial t}\lambda(t,r)}{r}$$

$$G_{1\,0} = \frac{\frac{\partial}{\partial t}\lambda(t,r)}{r}$$

$$G_{1\,1} = \frac{r\frac{\partial}{\partial r}\nu(t,r) - e^{\lambda(t,r)} + 1}{r^2}$$

$$G_{2\,2} = -\frac{r\left(re^{\lambda(t,r)}\left(\frac{\partial}{\partial t}\lambda(t,r)\right)^2 - re^{\lambda(t,r)}\frac{\partial}{\partial t}\lambda(t,r)\frac{\partial}{\partial t}\nu(t,r) + 2re^{\lambda(t,r)}\frac{\partial^2}{\partial t^2}\lambda(t,r) + re^{\nu(t,r)}\frac{\partial}{\partial r}\lambda(t,r)\frac{\partial}{\partial r}\nu(t,r) - r}{4}$$

$$G_{3\,3} = -\frac{r\left(re^{\lambda(t,r)}\left(\frac{\partial}{\partial t}\lambda(t,r)\right)^2 - re^{\lambda(t,r)}\frac{\partial}{\partial t}\lambda(t,r)\frac{\partial}{\partial t}\nu(t,r) + 2re^{\lambda(t,r)}\frac{\partial^2}{\partial t^2}\lambda(t,r) + re^{\nu(t,r)}\frac{\partial}{\partial r}\lambda(t,r)\frac{\partial}{\partial r}\nu(t,r) - r}{4}$$

I could not get it to calculate $G_\mu{}^\nu$, so I calculated manually.

$$G_0{}^0 = G_{0\,0}\cdot g^{00}$$

$$= \frac{\left(r\frac{\partial}{\partial r}\lambda(t,r) + e^{\lambda(t,r)} - 1\right)e^{-\lambda(t,r)}\cancel{e^{\nu(t,r)}}}{r^2}\cancel{e^{-\nu(t,r)}} \qquad \text{Sign error}$$

$$G_0{}^1 = G_{0\,1}g^{11}$$

$$= -\frac{\frac{\partial}{\partial t}\lambda(t,r)}{r}e^{-\lambda} \qquad \text{Pass}$$

$$G_1{}^1 = G_{1\,1}g^{11}$$

$$= -\frac{r\frac{\partial}{\partial r}\nu(t,r) - e^{\lambda(t,r)} + 1}{r^2}e^{-\lambda}$$

$$= -\frac{r\frac{\partial}{\partial r}\nu(t,r) - e^{\lambda(t,r)} + 1}{r^2}e^{-\lambda}$$

$$= \frac{1}{r^2} - e^{-\lambda}\frac{r\frac{\partial}{\partial r}\nu + 1}{r^2} \qquad \text{Pass}$$

References

[1] Marco San Martín and Joaquin Sureda. Pytearcat: Python tensor algebra calculator. *Astronomy & Astrophysics manuscript*, 2021.

[2] Emil T Akhmedov. Lectures on general theory of relativity. *arXiv preprint arXiv:1601.04996*, 2016.